SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: MLA03 COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Simulation-Based Assessment Question

Duration: 1hr

1. Simulate a **Hierarchical Reinforcement Learning (HRL)** agent for autonomous warehouse navigation.

(5 Marks)

2. Implement a **Hierarchical Abstract Machine (HAM)** to control a robot performing sequential tasks.

(5 Marks)

3. Develop the **MAXQ Value Decomposition** algorithm for a taxi navigation environment.

(5 Marks)

4. Simulate a **Meta-Reinforcement Learning** agent that rapidly adapts to multiple GridWorld environments.

(5 Marks)

5. Design a **Multi-Agent Reinforcement Learning (MARL)** system for cooperative robot path planning.

(5 Marks)

6. Implement a **competitive Multi-Agent RL** environment for autonomous drone navigation.

(5 Marks)

7. Simulate a **Partially Observable Markov Decision Process (POMDP)** for autonomous vehicle navigation under limited visibility.

(5 Marks)

8. Design an **ethical Reinforcement Learning** simulation for autonomous decisionmaking while satisfying safety constraints.

(5 Marks)

9. Develop an RL-based **adaptive traffic signal control** system and evaluate its performance.

10. Simulate an RL-based **smart energy management system** for optimizing power distribution in a microgrid.. **.**

(5 Marks)

ANSWER

1. Hierarchical Reinforcement Learning (HRL) for Autonomous Warehouse Navigation

Objective

To simulate an HRL agent that navigates an autonomous warehouse robot from a starting point to a target location while avoiding obstacles.

Concept

HRL divides a complex task into smaller subtasks.

High-level policy:

- Select target area/waypoint.

Low-level policy:

- Move the robot toward the selected waypoint.

Environment

A grid represents the warehouse:

- 0 → Free path

- 1 → Obstacle
- 2 → Target

Method

The high-level agent selects a subgoal such as **Move to Shelf A**, while the low-level agent chooses actions such as:

- Up
- Down
- Left
- Right

Reward

- +100 → Reach final destination
- +20 → Reach subgoal
- -10 → Hit obstacle
- -1 → Each movement

Expected Result

The HRL agent learns a hierarchical navigation strategy and reaches the destination with fewer unnecessary movements.

2. Hierarchical Abstract Machine (HAM) for Robot Sequential Tasks

Objective

To implement a Hierarchical Abstract Machine that controls a robot performing multiple tasks sequentially.

Example Tasks

The robot must:

Start → Pick Object → Move → Place Object → Return

HAM Structure

START

|

v

Move to Object

|

v

Pick Object

|

v

Move to Destination

|

v

Place Object

|

v

Return

Concept

HAM represents the task using hierarchical states and actions. Higher-level states control the overall task, while lower-level states perform individual actions.

Actions

- MOVE_FORWARD
- MOVE_LEFT
- MOVE_RIGHT
- PICK
- PLACE
- RETURN

Expected Result

The robot completes multiple sequential tasks according to the predefined hierarchical control structure.

Advantage

HAM reduces the complexity of learning by restricting the agent to valid actions for each task.

3. MAXQ Value Decomposition for Taxi Navigation

Objective

To implement the MAXQ hierarchical reinforcement learning algorithm in a taxi navigation environment.

Environment

The taxi must:

1. Reach passenger.
2. Pick up passenger.
3. Navigate to destination.
4. Drop passenger.

MAXQ Decomposition

MAXQ-Taxi

│

├── Navigate-to-Passenger

│

├── Pickup

│

├── Navigate-to-Destination

│

└── Dropoff

Concept

MAXQ decomposes the overall task into smaller subtasks and learns values for each subtask.

The overall value can be represented as:

[
Q_{MAXQ}(s,a)=Q_{subtask}(s,a)+C(s,a)
]

where:

- Qsubtask = value of the subtask
- C = completion value of the remaining tasks

Reward

Event	**Reward**
Successful pickup	+10
Successful drop-off	+100
Each movement	-1

Event	Reward
Invalid pickup/drop	-10

Expected Result

The taxi learns to efficiently pick up the passenger and deliver them to the correct destination.

4. Meta-Reinforcement Learning for Multiple GridWorld Environments

Objective

To simulate an RL agent that can rapidly adapt to different GridWorld environments.

Environments

For example:

- **Environment A:** Open grid
- **Environment B:** Obstacles
- **Environment C:** Different goal position
- **Environment D:** Different reward structure

Concept

Traditional RL learns one environment at a time.

Meta-RL attempts to learn **how to learn**, allowing the agent to adapt quickly to a new environment.

Training Environments

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Meta-RL Training

↓

Learn General Strategy

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New GridWorld

↓

Rapid Adaptation

Evaluation

Compare:

- Normal RL learning speed

- Meta-RL adaptation speed

Expected Result

The meta-RL agent should require fewer interactions to achieve good performance in a previously unseen GridWorld.

5. Multi-Agent RL for Cooperative Robot Path Planning

Objective

To design multiple cooperative robots that learn collision-free paths to their respective targets.

Environment

Example:

R1 → → → Target 1

↑

R2 → → → Target 2

Each robot is an RL agent.

Agents

- Robot 1
- Robot 2
- Robot 3

Actions

Each robot can:

- Up
- Down
- Left
- Right
- Stay

Reward

- +100 → Reach target
- -20 → Collision
- -1 → Movement
- +10 → Maintain safe distance

Cooperative Objective

The robots share information or rewards to improve the overall system performance.

Expected Result

Multiple robots learn collision-free paths and reach their targets efficiently.

Performance Metrics

- Average travel distance
- Number of collisions
- Completion time
- Total reward

6. Competitive Multi-Agent RL for Autonomous Drone Navigation

Objective

To simulate multiple autonomous drones competing for limited resources or navigation objectives.

Environment

Each drone attempts to reach a target while competing with other drones.

Drone 1 ———→ Target

↘

Competition

↗

Drone 2 ———→ Target

Actions

- Move up
- Move down
- Move left
- Move right
- Stay

Reward

Example:

- +100 → Reach target first
- +20 → Reach intermediate checkpoint
- -20 → Collision

- -1 → Each movement
- -50 → Enter restricted area

Competitive Learning

Unlike cooperative MARL, agents do not necessarily share the same objective.

The success of one drone may reduce the opportunity available to another.

Expected Result

Drones learn competitive navigation strategies while avoiding collisions and restricted zones.

7. POMDP for Autonomous Vehicle Navigation Under Limited Visibility

Objective

To simulate an autonomous vehicle navigating when it cannot observe the complete environment.

POMDP Components

A POMDP is represented by:

[
(S,A,T,R,\Omega,O)
]

where:

- S = States
- A = Actions
- T = Transition probabilities
- R = Rewards
- Ω = Observations
- O = Observation probabilities

Example

The vehicle may not know whether an obstacle is behind fog.

Vehicle → → → Fog → ?

|

Hidden Object

Actions

- Accelerate
- Brake

- Turn left
- Turn right
- Maintain speed

Observations

The vehicle receives incomplete information from sensors.

Reward

- +100 → Safely reach destination
- -100 → Collision
- -5 → Excessive braking
- -1 → Each time step

Expected Result

The vehicle learns safe navigation strategies despite incomplete environmental information.

8. Ethical Reinforcement Learning with Safety Constraints

Objective

To design an RL agent that maximizes rewards while satisfying ethical and safety constraints.

Example

An autonomous vehicle must reach its destination without:

- Hitting pedestrians
- Entering restricted zones
- Exceeding speed limits
- Causing collisions

Reward Function

A simple constrained reward can be:

[
R = R_{task} - \lambda R_{risk}
]

where:

- Rtask = task reward
- Rrisk = safety violation penalty
- λ = safety penalty coefficient

Safety Constraints

Collision = Not Allowed

Speed > Limit = Not Allowed

Restricted Area = Not Allowed

Pedestrian Risk = Minimize

Example Rewards

Event	Reward
Reach destination	+100
Safe movement	+5
Collision	-100
Speed violation	-30
Restricted zone	-50

Expected Result

The agent learns to achieve its objective while prioritizing safety and ethical constraints.

9. RL-Based Adaptive Traffic Signal Control

Objective

To develop an RL system that dynamically controls traffic signals to reduce congestion and waiting time.

State

The state can contain:

- Number of vehicles
- Queue length
- Current signal
- Waiting time
- Traffic density

Actions

Action 1 → Keep current signal

Action 2 → Change signal

Reward

A simple reward is:

[
R = -(\text{Queue Length}+\text{Waiting Time})
]

Lower congestion produces a higher reward.

RL Process

Traffic Environment

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State

↓

RL Agent

↓

Action

↓

Traffic Signal

↓

New Traffic State

↓

Reward

↓

Learning

Evaluation

Compare:

Fixed-time signal vs RL-based signal

Metrics:

- Average waiting time
- Average queue length
- Vehicle throughput
- Number of stops
- Total reward

Expected Result

The RL controller dynamically changes signal timing based on traffic conditions and reduces congestion compared with fixed-time control.

10. RL-Based Smart Energy Management for Microgrid

Objective

To simulate an RL agent that optimizes power distribution in a smart microgrid.

Environment Components

- Solar panels
- Wind energy
- Battery storage
- Grid
- Household/industrial loads

```
Solar ──────────┐
Wind ───────────┤
    ↓
  Microgrid
    ↓
 RL Controller
 ↙  ↓  ↘
Battery Load  Grid
```

State

The agent observes:

- Current energy demand
- Solar generation
- Wind generation
- Battery state of charge

- Electricity price
- Grid availability

Actions

- Charge battery
- Discharge battery
- Use renewable energy
- Purchase energy from grid
- Sell excess energy

Reward

A possible reward is:

[
R = -(\text{Energy Cost}+\text{Grid Dependency}+\text{Battery Degradation})
]

Objective

The agent attempts to:

- Minimize electricity cost.
- Maximize renewable energy utilization.
- Reduce unnecessary grid usage.
- Maintain battery health.
- Balance supply and demand.

Performance Metrics

- Total energy cost
- Renewable energy utilization
- Grid dependency
- Battery usage
- Peak demand
- Total reward

Expected Result

The RL agent learns an adaptive energy-management strategy that improves renewable-energy utilization and reduces operational cost.

Code of Conduct:

I __G.Srivardhan(192425208)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Srivardhan

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts

Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively